Q1 - 25 July - Shift 1

If $\alpha, \beta, \gamma, \delta$ are the roots of the equation $x^4 + x^3 + x^2 + x + 1 = 0$, then $\alpha^{2021} + \beta^{2021} + \gamma^{2021} + \delta^{2021}$ is equal to .

(A) -4 (B) -1

(C) 1 (D) 4

Space for your notes:

Q2 - 25 July - Shift 1

For $n \in N$, let $S_n = \left\{z \in C : |z - 3 + 2i| = \frac{n}{4}\right\}$ and $T_n = \left\{z \in C : |z - 2 + 3i| = \frac{1}{n}\right\}$.

Then the number of elements in the set $\{n \in N : S_n \cap T_n = \phi\}$ is :

(A) 0 (B) 2 (C) 3 (D) Infinite

Space for your notes:

Q3 - 25 July - Shift 2

For $z \in \mathbb{C}$ if the minimum value of $\left(|z - 3\sqrt{2}| + |z - p\sqrt{2}i|\right)$ is $5\sqrt{2}$, then a value of p is ______________

(A) 3 (B) $\frac{7}{2}$

(C) 4 (D) $\frac{9}{2}$

Space for your notes:

Q4 - 26 July - Shift 1

Let O be the origin and A be the point $z_1 = 1 + 2i$. If B is the point z_2, $Re(z_2) < 0$, such that OAB is a right angled isosceles triangle with OB as hypotenuse, then which of the following is NOT true ?

(A) $\arg z_2 = \pi - \tan^{-1} 3$

(B) $\arg (z_1 - 2z_2) = -\tan^{-1} \frac{4}{3}$

(C) $|z_2| = \sqrt{10}$

(D) $|2z_1 - z_2| = 5$

Space for your notes:

Q5 - 26 July - Shift 2

If $z = x + iy$ satisfies $|z| - 2 = 0$ and $|z-i|-|z+5i|=0$, then

(A) $x + 2y - 4 = 0$ (B) $x^2 + y - 4 = 0$

(C) $x + 2y + 4 = 0$ (D) $x^2 - y + 3 = 0$

Space for your notes:

Q6 - 27 July - Shift 1

Let the minimum value v_0 of $v = |z|^2 + |z-3|^2 + |z-6i|^2$, $z \in \mathbb{C}$ is attained at $z = z_0$. Then $\left|2z_0^2 - \bar{z}_0^3 + 3\right|^2 + v_0^2$ is equal to

(A) 1000 (B) 1024

(C) 1105 (D) 1196

Space for your notes:

Q7 - 27 July - Shift 1

Let $S=\{z\in\mathbb{C}:z^2+\bar{z}=0\}$. Then $\sum_{z\in S}(\operatorname{Re}(z)+\operatorname{Im}(z))$ is equal to ______.

Space for your notes:

Q8 - 27 July - Shift 2

Let S be the set of all (α,β), $\pi<\alpha,\beta<2\pi$, for which the complex number $\frac{1-i\sin\alpha}{1+2i\sin\alpha}$ is purely imaginary and $\frac{1+i\cos\beta}{1-2i\cos\beta}$ is purely real. Let $Z_{\alpha\beta}=\sin 2\alpha+i\cos 2\beta,(\alpha,\beta)\in S$.

Then $\sum_{(\alpha,\beta)\in S}\left(iZ_{\alpha\beta}+\frac{1}{i\bar{Z}_{\alpha\beta}}\right)$ is equal to :

(A) 3 (B) 3 i

(C) 1 (D) 2 – i

Space for your notes:

Q9 - 28 July - Shift 1

Let $S_1=\left\{z_1\in C:|z_1-3|=\frac{1}{2}\right\}$ and $S_2=\{z_2\in C:|z_2-|z_2+1||=|z_2+|z_2-1||\}$. Then, for $z_1\in S_1$ and $z_2\in S_2$, the least value of $|z_2-z_1|$ is :

(A) 0 (B) $\frac{1}{2}$ (C) $\frac{3}{2}$ (D) $\frac{5}{2}$

Space for your notes:

Q10 - 28 July - Shift 2

Let $z = a + ib$, $b \neq 0$ be complex numbers satisfying $z^2 = \bar{z} \cdot 2^{1-|z|}$. Then the least value of $n \in \mathbb{N}$, such that $z^n = (z+1)^n$, is equal to _____

Space for your notes:

Q11 - 29 July - Shift 1

If $z = 2 + 3i$, then $z^5 + (\bar{z})^5$ is equal to :

(A) 244
(B) 224
(C) 245
(D) 265

Space for your notes:

Q12 - 29 July - Shift 2

If $z \neq 0$ be a complex number such that $\left|z - \frac{1}{z}\right| = 2$, then the maximum value of $|z|$ is:

(A) $\sqrt{2}$
(B) 1
(C) $\sqrt{2} - 1$
(D) $\sqrt{2} + 1$

Space for your notes:

Q13 - 29 July - Shift 2

Let $S = \{z = x + iy : |z - 1 + i| \geq |z|, |z| < 2, |z + i| = |z - 1|\}$. Then the set of all values of x, for which $w = 2x + iy \in S$ for some $y \in \mathbb{R}$, is

(A) $\left(-\sqrt{2}, \frac{1}{2\sqrt{2}}\right]$
(B) $\left(-\frac{1}{\sqrt{2}}, \frac{1}{4}\right]$
(C) $\left(-\sqrt{2}, \frac{1}{2}\right]$
(D) $\left(-\frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}}\right]$

Space for your notes:

Answer Key

Q1 (B)	**Q2** (D)	**Q3** (C)	**Q4** (D)
Q5 (C)	**Q6** (A)	**Q7** (0)	**Q8** (C)
Q9 (C)	**Q10** (6)	**Q11** (A)	**Q12** (D)
Q13 (B)			

Q1 (B)

$\alpha, \beta, \gamma, \delta$ root of the equation

$x^4 + x^3 + x^2 + x + 1 = 0$

Which are 5^{th} roots of unity except 1.

then $\alpha^{2021} + \beta^{2021} + \gamma^{2021} + \delta^{2021} =$

$\alpha + \beta + \gamma + \delta = -1$

Q2 (D)

$S_n : |z - (3 - 2i)| = \frac{n}{4}$ is a circle center $C_1(3, -2)$

and radius n/4

$T_n : |z - (2 - 3i)| = \frac{1}{n}$ is a circle center C_2 (2,–3)

and radius 1/n

Here $S_n \cap T_n = \phi$

Both circles do not intersect each other

Case-1 : $C_1C_2 > n/4 + 1/n$

$\sqrt{2} > \frac{n}{4} + \frac{1}{n}$

then n = 1, 2, 3, 4

Case-2 : $C_1C_2 < \left|\frac{n}{4} - \frac{1}{n}\right|$

$\Rightarrow \sqrt{2} < \left|\frac{n^2 - 4}{4n}\right|$

$\Rightarrow$ n has infinite solutions for $n \in \mathbf{N}$

Q3 (C)

(C)

Q4 (D)

$$AB = AO.\ z^{-i\pi/2} = -2 + i$$

$$\text{So } OB = (-2 + i) + (1 + 2i)$$

$$z_2 = -1 + 3i$$

$$\therefore |2z_1 - z_2| = \sqrt{10}$$

Q5 (C)

$|z-i|-|z+5i|=0$

$\Rightarrow |x+(y-1)i| = |x+(y+5)i|$

$x^2+(y-1)^2 = x^2+(y+5)^2$

$(y-1)^2-(y+5)^2=0$

$(2y+4)(-6)=0$

$y=-2$

$\therefore\ x^2+(-2)^2=4$

$x=0$

$Z \equiv (0,-2)$, check options

Q6 (A)

$$z_0=\left(\frac{0+3+0}{3},\frac{0+6+0}{3}\right)=(1,2)$$

$$v_0=|1+2i|^2+|1+2i-3|^2+|1+2i-6i|^2=30$$

Then $\left|2z_0^2-\bar{z}_0^3+3\right|^2+v_0^2$

$$=\left|2(1+2i)^2-(1-2i)^3+3\right|^2+900$$

$$=|2(1-4+4i)-(1-4-4i)(1-2i)+3|^2+900$$

$$=|8+6i|^2+900=100+900=1000$$

Q7 (0)

$S=\{z\in C: z^2+\bar{z}=0\}$

Let $z = x + iy$

$z^2 = x^2 - y^2 + 2ixy$

$\bar{z} = x - iy$

$z^2+\bar{z} = x^2 - y^2 + x + i(2xy - y) = 0$

$\Rightarrow x^2 + x - y^2 = 0 \ \& \ 2xy - y = 0$

$y = 0$ or $x = \frac{1}{2}$

If $y = 0$; $x = 0, -1$

If $x = \frac{1}{2}; y = \frac{\sqrt{3}}{2}, \frac{-\sqrt{3}}{2}$

$$\sum_{z\in S}\left(\mathrm{Re}(z)+\mathrm{Im}(z)=\left(0-1+\frac{1}{2}+\frac{1}{2}\right)+0+0+\frac{\sqrt{3}}{2}-\frac{\sqrt{3}}{2}\right)$$

Q8 (C)

$\pi < \alpha, \beta < 2\pi$

$\frac{1 - i\sin\alpha}{1 + i(2\sin\alpha)}$ = Purely imaginary

$\Rightarrow \frac{(1 - i\sin\alpha)(1 - i(2\sin\alpha))}{1 + 4\sin^2\alpha}$ = Purely imaginary

$\Rightarrow \frac{1 - 2\sin^2\alpha}{1 + 4\sin^2\alpha} = 0$

$\Rightarrow \sin^2\alpha = \frac{1}{2}$

$\Rightarrow \alpha = \left\{\frac{5\pi}{4}, \frac{7\pi}{4}\right\}$

& $\frac{1 + i\cos\beta}{1 + i(-2\cos\beta)}$ = Purely real

$\Rightarrow \frac{(1 + i\cos\beta)(1 + 2i\cos\beta)}{1 + 4\cos^2\beta}$ = Purely real

$\Rightarrow 3\cos\beta = 0$

$\Rightarrow \boxed{\beta = \frac{3\pi}{2}}$

$\Rightarrow Z_{\alpha\beta} = \sin\frac{5\pi}{2} + i\cos 3\pi = 1 - i$

or

$Z_{\alpha\beta} = \sin\frac{7\pi}{2} + i\cos 3\pi = -1 - i$

Required value = $\left[i(1-i) + \frac{1}{i(1+i)}\right] + \left[i(-1-i) + \frac{1}{i(-1+i)}\right]$

$= i(-2i) + \frac{1}{i}\frac{2i}{(-2)} \Rightarrow 2 - 1 = 1$

Q9 (C)

$$\left|z_2+|z_2-1|\right|^2=\left|z_2-|z_2+1|\right|^2$$

$$\Rightarrow \left|z_2+|z_2-1|\right|\left(\bar{z}_2+|z_2-1|\right)=\left(z_2-|z_2+1|\right)\left(\bar{z}_2-(z_2+1)\right)$$

$$\Rightarrow \quad z_2|\bar{z}_2+12_2-1|-\left(\bar{z}_2-|z_2+1|\right)+\bar{z}_2\left(|z_2-1|+|z_2+1|\right)$$

$$=|z_2+1|^2=|z_2-1|^2$$

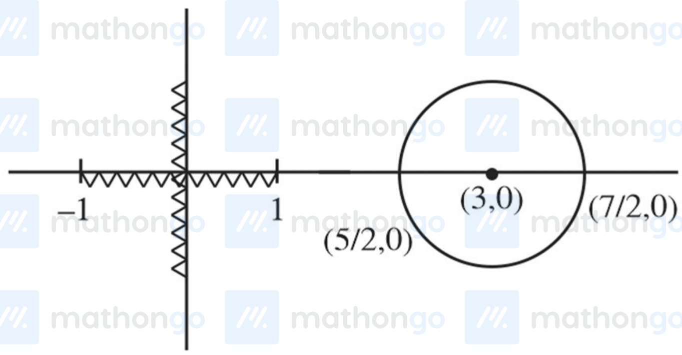
$$\Rightarrow [z_2+\bar{z}_2)(|z_2-1|)+(z_2+1|)=2(z_2+\bar{z}_2)$$

$$\Rightarrow (z_2+\bar{z}_2)(|z_2-1|+|z_2+1|-2)=0$$

$\therefore\ z_2+\bar{z}_2=0$ or $|z_2-1|+|z_2+1|-2=0$

$\therefore\ z_2$ lie on imaginary axis. Or on real axis with in $[-1,1]$

Also $|z_1-3|=\dfrac{1}{2}$ lie on circle having centre 3 and radius $\dfrac{1}{2}$.



Clearly $|z_1-z_2|\min=\dfrac{5}{2}-1=\dfrac{3}{2}$

Q10 (6)

$$\left|z^2\right| = |\bar{z}| \cdot 2^{1-|z|} \Rightarrow |z| = 1$$

$$z^2 = \bar{z} \Rightarrow z^3 = 1 \therefore z = \omega \text{ or } \omega^2$$

$$\omega^n = (1+\omega)^n = \left(-\omega^2\right)^n$$

Least natural value of n is 6.

Q11 (A)

$$z^5 + (\bar{z})^5 = (2+3i)^5 + (2-3i)^5$$

$$= 2({}^5C_0 2^5 + {}^5C_2 2^3 (3i)^2 + {}^5C_4 2^1 (3i)^4)$$

$$= 2\,(32 + 10 \times 8\,(-9) + 5 \times 2 \times 81) = 244$$

Q12 (D)

$|z - 1/z| = 2$

$$\left||z| - \frac{1}{|z|}\right| \le \left|z - \frac{1}{z}\right| \le |z| + \frac{1}{|z|} \qquad \text{Let } |z| = r$$

$$\left|r - \frac{1}{r}\right| \le 2 \le r + \frac{1}{r}$$

$$\left|r - \frac{1}{r}\right| \le 2 \;\&\; r + \frac{1}{r} \ge 2 \text{ always true}$$

$$r - \frac{1}{r} \ge -2 \;\&\; r - \frac{1}{r} \le 2$$

$r^2 - 1 \le 2r$

$r^2 - 2r \le 1$

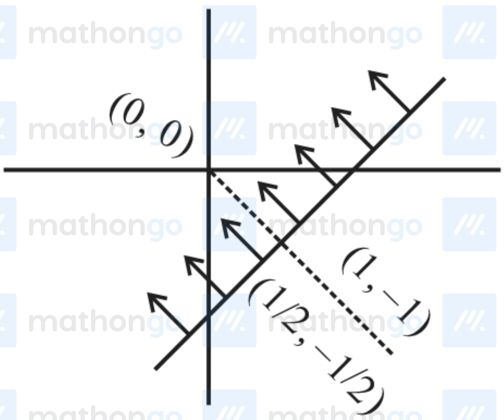
$(r - 1)^2 \le 2$

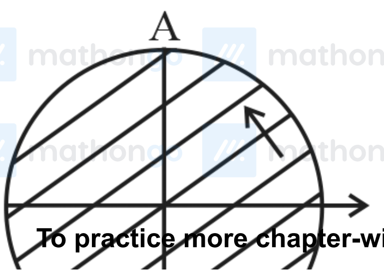
$r - 1 \le \sqrt{2}$

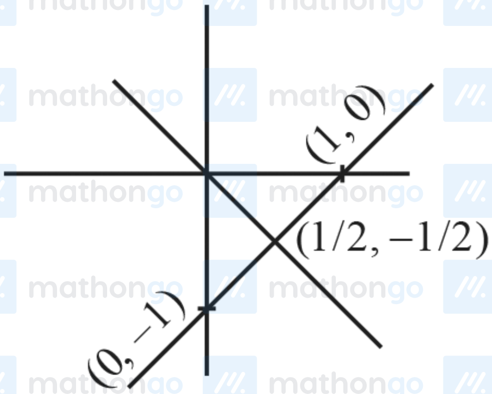
$\therefore |z|_{max} = 1 + \sqrt{2}$

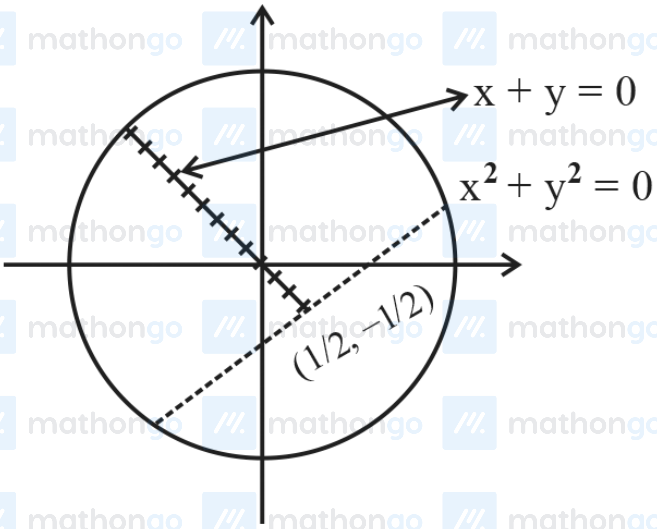
Q13 (B)

$|z - 1 + i| \ge |z|$; $|z| < 2$; $|z + i| = |z - 1|$









Hence

$w = 2x + iy \in S$

$2x \leq \frac{1}{2} \quad \therefore x \leq \frac{1}{4}$

Now

$(2x)^2 + (2x)^2 < 4$

$x^2 < \frac{1}{2} \Rightarrow x \in \left(\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

$\therefore x \in \left(\frac{-1}{\sqrt{2}}, \frac{1}{4}\right]$